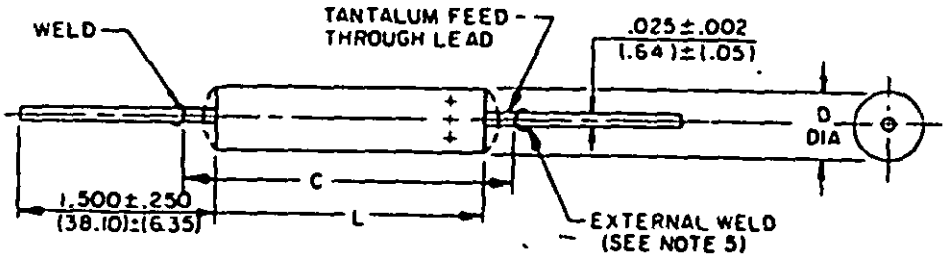
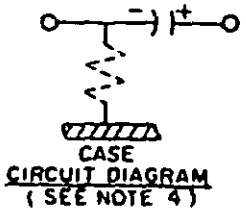
MILITARY SPECIFICATION SHEET

CAPACITORS, FIXED, ELECTROLYTIC (NONSOLID ELECTROLYTE), TANTALUM, (POLARIZED, ETCHED FOIL), 85°C (VOLTAGE DERATED TO 125°C), ESTABLISHED RELIABILITY, STYLE CLR25

The complete requirements for procuring the capacitors described herein shall consist of this document and the latest issue of Specification MIL-C-39006.

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.





© Case size	Dimensions			
	Basic case		Insulated case	
	L 1/ ±.062(1.57)	D ±.016(.41)	D max	C max
E1	.688(17.48)	.188(4.78)	.219 (5.56)	1.188(30.18)
G2	.969(24.61)	.281(7.14)	.312 (7.92)	1.469(37.31)
G3	1.438(36.53)	.375(9.53)	.406(10.31)	1.938(49.23)
G4	2.125(53.98)	.375(9.53)	.406(10.31)	2.625(66.68)
G5	2.750(69.85)	.375(9.53)	.406(10.31)	3.250(82.55)

1/ Length of basic case; sleeving shall be as specified in MIL-C-39006.

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are in parentheses.
3. Metric equivalents (to the nearest .01 mm) are given for general information only and are based upon 1 inch = 25.4 mm.
4. There is an indeterminate resistance between the metal case and the negative terminal.
5. The weld shall not be enclosed in the end seal.

FIGURE 1. Style CLR25 capacitors.

© denotes changes.

FSC 5910

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TABLE I. Capacitor characteristics.

Rated voltage (85° C)	Cap.	Cap. tolerance	DC leakage (max) 25° C	DC leakage (max) 85° C & 125° C	Derated voltage (125° C)	Surge voltage (85° C)	Impedance (max)	Case size	Part No. M39006/01- Failure rate level (%/1,000 hr) L(2.0)	M(1.0)	P(0.1)	R(0.01)	S(0.001)
HERMETICALLY-SEALED													
volts, dc	µF	%	µA	µA	volts, dc	volts, dc	ohms						
15	60.0	+75, -15	2	10	10	17.2	55	G2	3001	3030	3059	3088	3117
15	200.0	+75, -15	3	40	10	17.2	17	G3	3002	3031	3060	3089	3118
15	400.0	+75, -15	6	80	10	17.2	9	G4	3003	3032	3061	3090	3119
15	580.0	+75, -15	9	100	10	17.2	6	G5	3004	3033	3062	3091	3120
25	40.0	+75, -15	2	13	15	28.8	83	G2	3005	3034	3063	3092	3121
25	120.0	+75, -15	3	40	15	28.8	27	G3	3006	3035	3064	3093	3122
25	250.0	+75, -15	6	80	15	28.8	13	G4	3007	3036	3065	3094	3123
25	350.0	+75, -15	10	100	15	28.8	10	G5	3008	3037	3066	3095	3124
30	32.0	+75, -15	2	14	20	34.5	106	G2	3009	3038	3067	3096	3125
30	110.0	+75, -15	4	50	20	34.5	31	G3	3010	3039	3068	3097	3126
30	220.0	+75, -15	8	90	20	34.5	16	G4	3011	3040	3069	3098	3127
30	300.0	+75, -15	11	120	20	34.5	11	G5	3012	3041	3070	3099	3128
50	18.0	+50, -15	2	13	30	57.5	180	G2	3013	3042	3071	3100	3129
50	60.0	+50, -15	3	40	30	57.5	55	G3	3014	3043	3072	3101	3130
50	75.0	+50, -15	4	70	30	57.5	52	G5	3015	3044	3073	3102	3131
50	100.0	+50, -15	4	75	30	57.5	33	G4	3016	3045	3074	3103	3132
50	150.0	+50, -15	5	100	30	57.5	22	G5	3017	3046	3075	3104	3133
75	12.0	+50, -15	2	13	50	86.2	270	G2	3018	3047	3076	3105	3134
75	30.0	+50, -15	3	34	50	86.2	110	G3	3019	3048	3077	3106	3135
75	70.0	+50, -15	6	80	50	86.2	47	G4	3020	3049	3078	3107	3136
75	100.0	+50, -15	10	100	50	86.2	33	G5	3021	3050	3079	3108	3137
100	8.0	+30, -15	2	12	65	115.0	410	G2	3022	3051	3080	3109	3138
100	25.0	+30, -15	2.5	37	65	115.0	130	G3	3023	3052	3081	3110	3139
100	50.0	+30, -15	5	75	65	115.0	66	G4	3024	3053	3082	3111	3140
100	70.0	+30, -15	7	100	65	115.0	47	G5	3025	3054	3083	3112	3141
150	4.0	+30, -15	2	13	100	172.0	750	G2	3026	3055	3084	3113	3142
150	13.0	+30, -15	3	30	100	172.0	250	G3	3027	3056	3085	3114	3143
150	25.0	+30, -15	5	80	100	172.0	130	G4	3028	3057	3086	3115	3144
150	36.0	+30, -15	8	100	100	172.0	94	G5	3029	3058	3087	3116	3145
NONHERMETICALLY-SEALED													
15	15.0	+75, -15	2	4	10	17.2	220	E1	2200	2232	2264	2296	2328
25	10.0	+75, -15	2	4	15	28.8	330	E1	2205	2237	2269	2301	2333
30	8.0	+75, -15	2	6	20	34.5	424	E1	2210	2242	2274	2306	2338
50	4.5	+50, -15	2	4	30	57.5	730	E1	2212	2244	2276	2308	2340
75	3.0	+50, -15	2	4	50	86.2	1000	E1	2217	2249	2281	2313	2345
100	2.0	+30, -15	2	4	65	115.0	1600	E1	2222	2254	2286	2318	2350
150	1.0	+30, -15	2	4	100	172.0	3000	E1	2227	2259	2291	2323	2355

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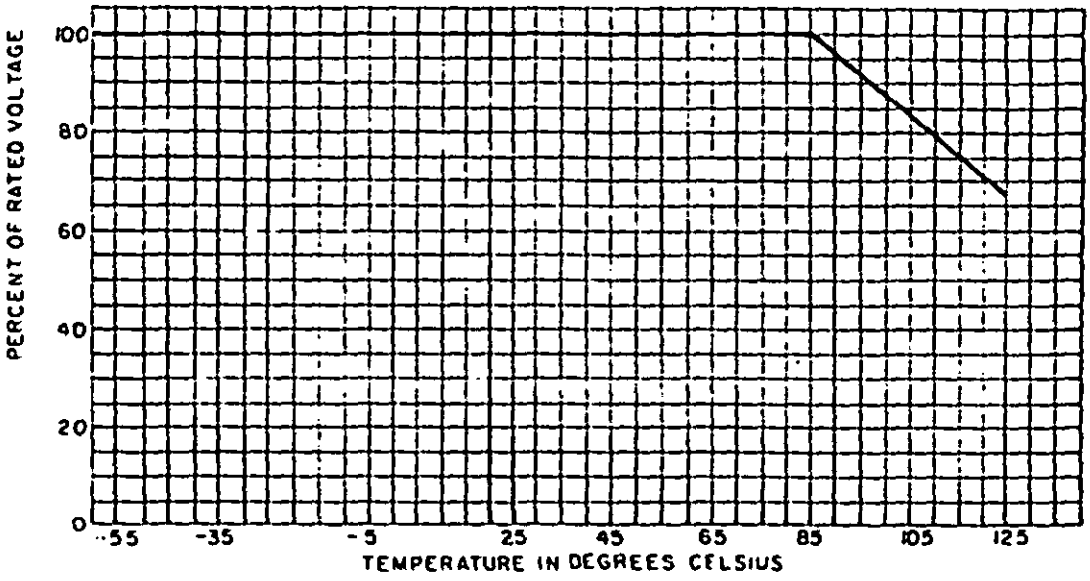


FIGURE 2. Voltage derating with temperature.

REQUIREMENTS:

Design and construction:
Dimensions and configuration - See figure 1.
Case type - Tubular, insulated.
Seal type - G (hermetic) or E (nonhermetic) as shown in table I.
Terminals - Axial-wire lead.
Rated temperature - -55° to +85°C, voltage derated to +125°C (see figure 2).
Rated voltage - See table I.
Capacitance (Cap.) (nom) - See table I.
Cap. tolerance - See table I.
FR level - See table I.

Constant voltage conditioning: As specified in MIL-C-39006.

DC leakage (DCL) (max) (at 25°C): See table I.

Cap. (nom): Method 305 of MIL-STD-202. See table I.

Dissipation factor (DF) (max):
50 volts and below - 20%.
Above 50 volts - 15%.

Seal (G seal type only): Method 112 of MIL-STD-202, conditions A and C.

Shock (specified pulse): Method 213 of MIL-STD-202, condition I (100G).

Vibration, high frequency: Method 204 of MIL-STD-202, condition D (20G).

Temperature cycling: Method 102 of MIL-STD-202, condition D (with step 3 at +125°C). Number of cycles - 30.
DCL - 125% (max) of 25°C value of table I.
ΔCap. - Within ±5% of initial measurement.
DF - 115% (max) of initial requirement.

Salt spray (corrosion): Method 101 of MIL-STD-202, condition B (48 hr).

Solderability: Method 208 of MIL-STD-202.

Terminal strength:
Pull - Method 211 of MIL-STD-202, condition A.
Wire-lead bend - In accordance with MIL-C-39006.

Surge voltage: - See table I.

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Moisture resistance: Method 106 of MIL-STD-202.
- DCL - 125% (max) of 25°C value of table I.
- ΔCap. - Within ±6% of initial measurement.
- DF - 115% (max) of initial requirement.

Dielectric withstanding voltage (DWV): Method 301 of MIL-STD-202.
- Test voltage - 2,000 vdc, min.

Insulation resistance (IR): Method 302 of MIL-STD-202, condition B. 100 megohms, min.

Low temperature (storage): Method 502 of MIL-STD-810.
- DCL - See table I.
- ΔCap. - Within ±5% of initial measurement.
- DF - 20% (≤50v) or 15% (>50v) (max).

Stability at low and high temperatures:
- Step 1 (+25°C):
  - DCL - See table I.
  - Cap. - Within tolerance of table I value.
  - DF - 20% (≤50v) or 15% (>50v) (max).
- Step 2 (-55°C):
  - Imp. - See table I.
  - ΔCap. - -40% (≤50v) or -35% (>50v) of step 1 value (max).
- Step 3 (+25°C):
  - DCL - See table I.
  - ΔCap. - Within ±5% of step 1 value.
  - DF - 20% (≤50v) or 15% (>50v) (max).
- Step 4 (+85°C):
  - DCL - See table I.
  - ΔCap. - +20% of step 1 value (max).
  - DF - 30% (≤50v) or 20% (>50v).
- 1/ Step 5 (+125°C):
  - DCL - See table I.
  - ΔCap. - +50% (≤30v) or +30% (>30v) of step 1 value (max).
  - DF - 30% (≤30v) or 20% (>30v) (max).
- Step 6 (+25°C):
  - DCL - See table I.
  - ΔCap. - Within ±5% of step 1 value.
  - DF - 20% (≤50v) or 15% (>50v) (max).

Reverse voltage:
- DCL - 125% (max) of 25°C value of table I.
- ΔCap. - Within ±10% of initial measurement.
- DF - 20% (≤50v) or 15% (>50v) (max).

Life at 85°C: Method 108 of MIL-STD-202.
- 2,000-hr (qualification):
  - DCL - See table I.
  - ΔCap. - Within ±15% of initial measurement.
  - DF - 20% (≤50v) or 15% (>50v) (max).
- 240-hr (group B):
  - DCL - See table I.
  - ΔCap. - Within ±10% of initial measurement.
  - DF - 20% (≤50v) or 15% (>50v) (max).
- 10,000-hr (group C) and extended life:
  - DCL (85°C) - 125% (max) of table I value.
  - DCL (25°C) - See table I.
  - ΔCap. - Within ±25% of initial measurement.
  - DF - 40% (≤50v) or 30% (>50v) (max).
  - DWV - 2,000 vdc, min.
  - IR - 100 megohms, min.

Vacuum life (G seal type only):
- DCL - See table I.
- ΔCap. - Within ±20% of initial measurement.
- DF - 20% (≤50v) or 15% (>50v) (max).
- Weight loss - 4.1 mg, max.

1/ Life at 125°C: Method 108 of MIL-STD-202.
- DCL - See table I.
- ΔCap. - Within ±15% of initial measurement.
- DF - 20% (≤30v) or 15% (>30v) (max).
- DWV - 2,000 vdc, min.
- IR - 100 megohms, min.

Barometric pressure: Method 105 of MIL-STD-202, condition E (150,000 ft).
- Test voltage - Rated voltage.

Fungus: Method 508 of MIL-STD-810.

Resistance to solvents: Method 215 of MIL-STD-202.

Resistance to soldering heat: Method 210 of MIL-STD-202, condition B.
- DCL - See table I.
- ΔCap. - Within ±5% of initial measurement.
- DF - 20% (≤50v) or 15% (>50v) (max).

Marking (except case size E1): In accordance with MIL-C-39006. E1 case size, mark as follows (the trademark may also be placed on the unit where space permits):

Example:

M39006/ 01-2200	- Part number.
J12345	- "J" for "JAN" brand and source code.
7144A	- Date code and lot symbol.
15μF 15V	- Capacitance and rated voltage.

Polarity - Series of plus (+) symbols encircling positive end (see figure 1).

1/ 125°C derated voltage referenced.

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TABLE II. Substitutability data - "C" seal type.

Dash no. in MIL-C-39006/1C and /1H					Substitutable for dash no. in: 1/ MIL-C-39006/1A					2/ MIL-C-39006/1					3/ MIL-C-39006/5				
L	M	P	R	S	L	M	P	R	S	L	M	P	R	S	L	M	P	R	S
3001	3030	3059	3088	3117	1001	1027	1053	1079	1105	0006	0007	0008	0009	0010			None		
3002	3031	3060	3089	3118	1002	1028	1054	1080	1106	0011	0012	0013	0014	0015			None		
3003	3032	3061	3090	3119	1003	1029	1055	1081	1107	0016	0017	0018	0019	0020			None		
3004	3033	3062	3091	3120	1004	1030	1056	1082	1108	0021	0022	0023	0024	0025	0011	0012	0013	0014	0015
3005	3034	3063	3092	3121	1005	1031	1057	1083	1109	0031	0032	0033	0034	0035			None		
3006	3035	3064	3093	3122	1006	1032	1058	1084	1110	0036	0037	0038	0039	0040			None		
3007	3036	3065	3094	3123	1007	1033	1059	1085	1111	0041	0042	0043	0044	0045	0006	0007	0008	0009	0010
3008	3037	3066	3095	3124	1008	1034	1060	1086	1112	0046	0047	0048	0049	0050			None		
3009	3038	3067	3096	3125			None					None					None		
3010	3039	3068	3097	3126			None					None					None		
3011	3040	3069	3098	3127			None					None					None		
3012	3041	3070	3099	3128			None					None					None		
3013	3042	3071	3100	3129	1009	1035	1061	1087	1113	0056	0057	0058	0059	0060	0031	0032	0033	0034	0035
3014	3043	3072	3101	3130	1010	1036	1062	1088	1114	0061	0062	0063	0064	0065	0036	0037	0038	0039	0040
3015	3044	4/3073	3102	3131	1011	1037	4/1063	1089	1115			None					None		
3016	3045	3074	3103	3132	1012	1038	1064	1090	1116	0066	0067	0068	0069	0070	0021	0022	0023	0024	0025
3017	3046	3075	3104	3133	1013	1039	1065	1091	1117	0071	0072	0073	0074	0075	0041	0042	0043	0044	0045
3018	3047	3076	3105	3134	1015	1041	1067	1093	1119	0081	0082	0083	0084	0085	0046	0047	0048	0049	0050
3019	3048	3077	3106	3135	1016	1042	1068	1094	1120	0086	0087	0088	0089	0090	0051	0052	0053	0054	0055
3020	3049	3078	3107	3136	1017	1043	1069	1095	1121	0091	0092	0093	0094	0095			None		
3021	3050	3079	3108	3137	1018	1044	1070	1096	1122	0096	0097	0098	0099	0100	0056	0057	0058	0059	0060
3022	3051	3080	3109	3138	1019	1045	1071	1097	1123	0106	0107	0108	0109	0110			None		
3023	3052	3081	3110	3139	1020	1046	1072	1098	1124	0111	0112	0113	0114	0115			None		
3024	3053	3082	3111	3140	1021	1047	1073	1099	1125	0116	0117	0118	0119	0120			None		
3025	3054	3083	3112	3141	1022	1048	1074	1100	1126	0121	0122	0123	0124	0125			None		
3026	3055	3084	3113	3142	1023	1049	1075	1101	1127	0131	0132	0133	0134	0135			None		
3027	3056	3085	3114	3143	1024	1050	1076	1102	1128	0136	0137	0138	0139	0140	0081	0082	0083	0084	0085
3028	3057	3086	3115	3144	1025	1051	1077	1103	1129	0141	0142	0143	0144	0145	0066	0067	0068	0069	0070
3029	3058	3087	3116	3145	1026	1052	1078	1104	1130	0146	0147	0148	0149	0150	0086	0087	0088	0089	0090

1/ Dash nos. 1014, 1040, 1066, 1092, and 1118 deleted with no replacement.

2/ Dash nos. 0001 thru 0005; 0026 thru 0030; 0051 thru 0055; 0076 thru 0080; 0101 thru 0105; and 0126 thru 0130 deleted with no replacement.

3/ Dash nos. 0001 thru 0005; 0016 thru 0020; 0026 thru 0030; 0061 thru 0065; and 0071 thru 0080 deleted with no replacement.

4/ Dash no. 3073 is substitutable for no. 1063 of MIL-C-39006/1A and type designation 7ECDA756TP of MIL-C-38102/7B (USAF).

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TABLE III. Substitutability data - "E" seal type.

Dash no. in MIL-C-39006/1C and /1B					Substitutable for dash no. in: MIL-C-39006/1B					MIL-C-39006/1A					MIL-C-39006/11				
L	M	P	R	S	L	M	P	R	S	L	M	P	R	S	L	M	P	R	S
3001	3030	3059	3088	3117	2201	2233	2265	2297	2329	2002	2032	2062	2092	2122	0006	0007	0008	0009	0010
3002	3031	3060	3089	3118	2202	2234	2266	2298	2330	2003	2033	2063	2093	2123	0011	0012	0013	0014	0015
3003	3032	3061	3090	3119	2203	2235	2267	2299	2331	2004	2034	2064	2094	2124	0016	0017	0018	0019	0020
3004	3033	3062	3091	3120	2204	2236	2268	2300	2332	2005	2035	2065	2095	2125	0021	0022	0023	0024	0025
3005	3034	3063	3092	3121	2206	2238	2270	2302	2334	2007	2037	2067	2097	2127	0031	0032	0033	0034	0035
3006	3035	3064	3093	3122	2207	2239	2271	2303	2335	2008	2038	2068	2098	2128	0036	0037	0038	0039	0040
3007	3036	3065	3094	3123	2208	2240	2272	2304	2336	2009	2039	2069	2099	2129	0041	0042	0043	0044	0045
3008	3037	3066	3095	3124	2209	2241	2273	2305	2337	2010	2040	2070	2100	2130	0046	0047	0048	0049	0050
3009	3038	3067	3096	3125	2211	2243	2275	2307	2339			None					None		
3010	3039	3068	3097	3126			None					None					None		
3011	3040	3069	3098	3127			None					None					None		
3012	3041	3070	3099	3128			None					None					None		
3013	3042	3071	3100	3129	2213	2245	2277	2309	2341	2012	2042	2072	2102	2132	0056	0057	0058	0059	0060
3014	3043	3072	3101	3130	2214	2246	2278	2310	2342	2013	2043	2073	2103	2133	0061	0062	0063	0064	0065
3015	3044	3073	3102	3131			None					None					None		
3016	3045	3074	3103	3132	2215	2247	2279	2311	2343	2014	2044	2074	2104	2134	0066	0067	0068	0069	0070
3017	3046	3075	3104	3133	2216	2248	2280	2312	2344	2015	2045	2075	2105	2135	0071	0072	0073	0074	0075
3018	3047	3076	3105	3134	2218	2250	2282	2314	2346	2017	2047	2077	2107	2137	0081	0082	0083	0084	0085
3019	3048	3077	3106	3135	2219	2251	2283	2315	2347	2018	2048	2078	2108	2138	0086	0087	0088	0089	0090
3020	3049	3078	3107	3136	2220	2252	2284	2316	2348	2019	2049	2079	2109	2139	0091	0092	0093	0094	0095
3021	3050	3079	3108	3137	2221	2253	2285	2317	2349	2020	2050	2080	2110	2140	0096	0097	0098	0099	0100
3022	3051	3080	3109	3138	2223	2255	2287	2319	2351	2022	2052	2082	2112	2142	0106	0107	0108	0109	0110
3023	3052	3081	3110	3139	2224	2256	2288	2320	2352	2023	2053	2083	2113	2143	0111	0112	0113	0114	0115
3024	3053	3082	3111	3140	2225	2257	2289	2321	2353	2024	2054	2084	2114	2144	0116	0117	0118	0119	0120
3025	3054	3083	3112	3141	2226	2258	2290	2322	2354	2025	2055	2085	2115	2145	0121	0122	0123	0124	0125
3026	3055	3084	3113	3142	2228	2260	2292	2324	2356	2027	2057	2087	2117	2147	0131	0132	0133	0134	0135
3027	3056	3085	3114	3143	2229	2261	2293	2325	2357	2028	2058	2088	2118	2148	0136	0137	0138	0139	0140
3028	3057	3086	3115	3144	2230	2262	2294	2326	2358	2029	2059	2089	2119	2149	0141	0142	0143	0144	0145
3029	3058	3087	3116	3145	2231	2263	2295	2327	2359	2030	2060	2090	2120	2150	0146	0147	0148	0149	0150

Dash no. in MIL-C-39006/1C and /1B					Substitutable for dash no. in MIL-C-39006/1A					MIL-C-39006/11				
L	M	P	R	S	L	M	P	R	S	L	M	P	R	S
2200	2232	2264	2296	2328	2001	2031	2061	2091	2121	0001	0002	0003	0004	0005
2205	2237	2269	2301	2333	2006	2036	2066	2096	2126	0026	0027	0028	0029	0030
2210	2242	2274	2306	2338			None					None		
2212	2244	2276	2308	2340	2011	2041	2071	2101	2131	0051	0052	0053	0054	0055
2217	2249	2281	2313	2345	2016	2046	2076	2106	2136	0076	0077	0078	0079	0080
2222	2254	2286	2318	2350	2021	2051	2081	2111	2141	0101	0102	0103	0104	0105
2227	2259	2291	2323	2355	2026	2056	2086	2116	2146	0126	0127	0128	0129	0130

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Voltage groups (85°C rated voltage):
I - - - - - - - - - - - 15 to 100v incl
II - - - - - - - - - - - 150v

NOTE: The capacitors covered by this specification sheet are considered substitutable with the capacitors listed in table II or III.

Custodians:
Army - EL
Navy - EC
Air Force - 11

Review activities:
Army - MU
Navy - SH
Air Force - 17, 80
DSA - ES

User activities:
Army -
Navy - AS, MC, OS
Air Force - 19

Preparing activity:
Navy - EC

Agent:
DSA - ES

(Project 5910-1055-1)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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